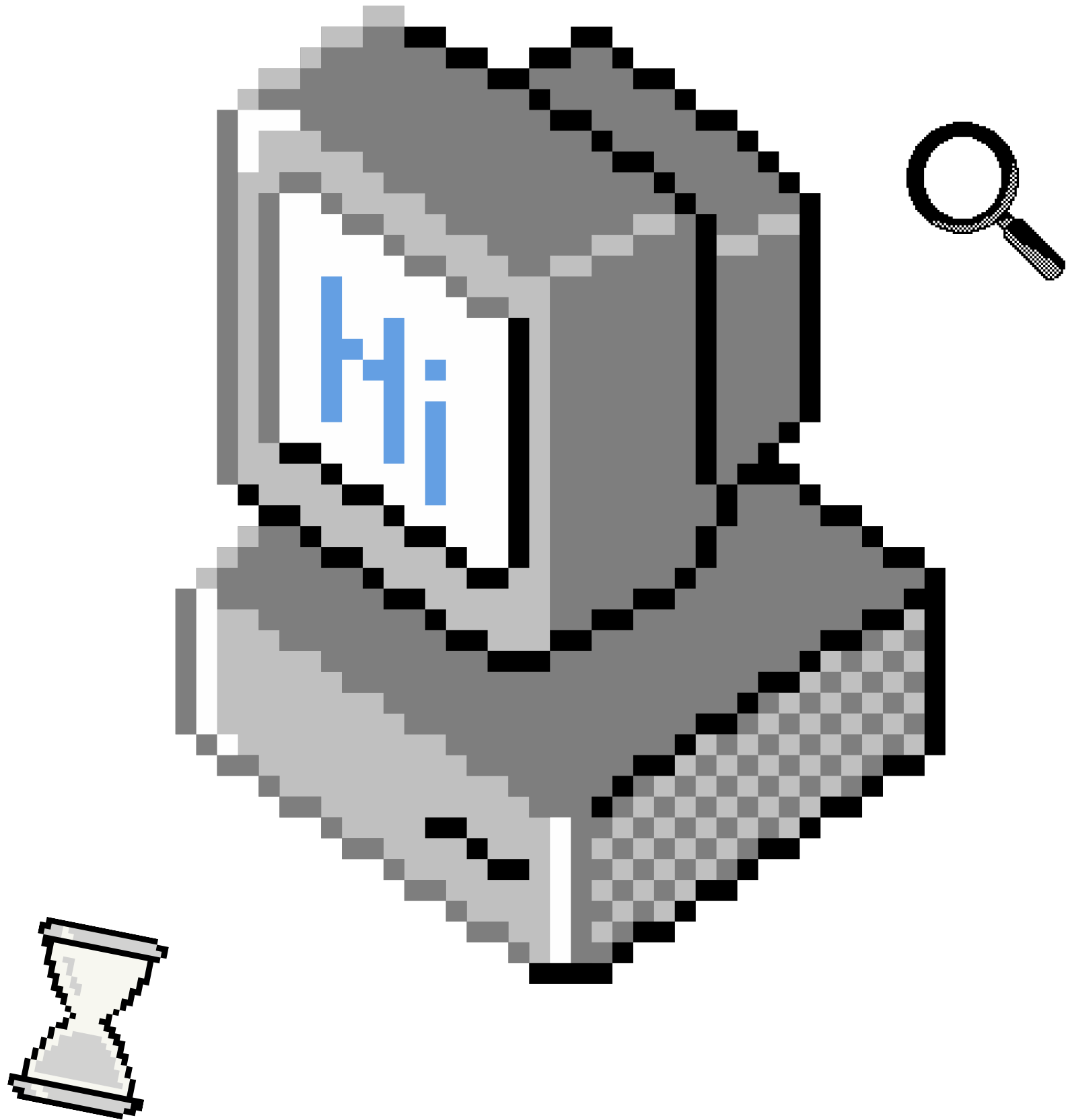


MACHINE BEHAVIOR AND HUMAN ACTION



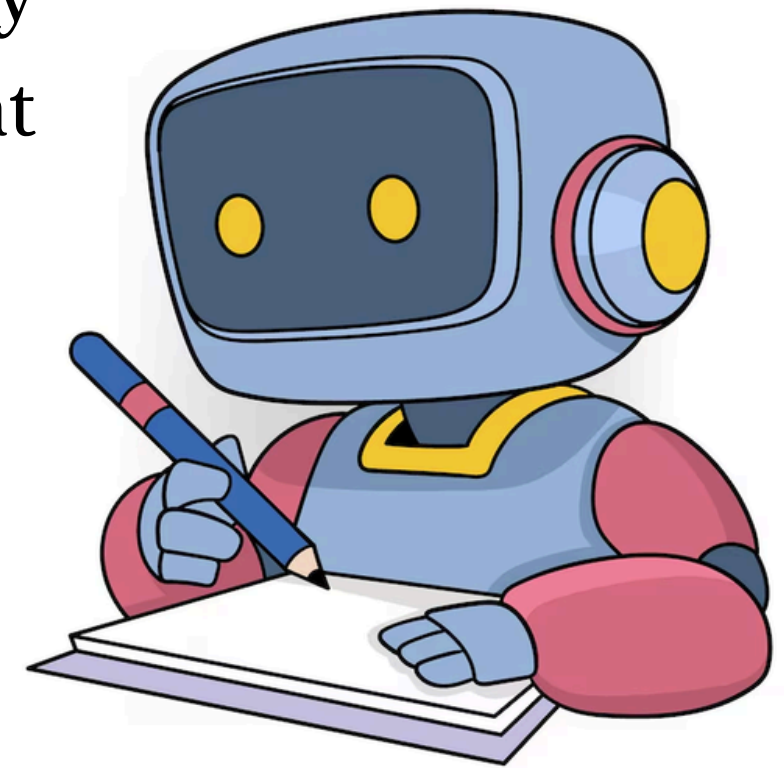
Shape of actions

WRITING MACHINES

- Speech transcribes
- lettering
- The signature

→ A Escrita envolve múltiplos níveis de ação, nem todos são automatizáveis

The polymorphic actions, which cannot be mimicked by machines without alteration of the action, and the mimeomorphic actions, which can, are deeply intermingled in writing. In the case of writing, we tend to think that mimeomorphicity goes higher up the action tree than it does (p. 115)



A ESCRITA É UMA AÇÃO COMPLEXA

Page 03

Lettering

we do not admire people for their ability to use different fonts, whereas we do applaud or admire the versatile calligrapher. Of course, there is a residue of polymorphic action to be admired in the creative calligrapher, but the point we are making here is that even the mimeomorphic element of calligraphy can be admired when executed by a skillful human.

The signature

The signature machine, then, fulfills the same role as an official seal, rather than that of a human being. In the case of junk mail, it is rather hard to see what the printed signature is for; perhaps some take it as a symbol of personal attention, but surely only the most gullible. In the case of a typed letter, the sender is expected to sign by hand if they mean to show that it has been composed—or at least read—by the sender.

We see here a machine used to effect a deceit—a case of an inauthentically executed mimeomorphic action. It is interesting that even mimeomorphic actions can be inauthentic. Thus we see that the distinction between authenticity and inauthenticity does not map onto the distinction between polymorphic and mimeomorphic actions

Spell checkers and Bibliography checkers

The spell checker would have to understand the whole sense of this passage if it were to spell as well as the typical editor. Thus we value a degree of polymorphicity in our spelling, and because of this it is currently impossible to design a spell checker that checks in the same way a human checks

we would predict that referencing conventions will converge and that eventually referencing that is not aided by machine will become rare.

WHAT MACHINES CAN DO

Page 04

Tools

Machines can amplify our ability to do what we can already do—let us call such machines “tools.”

A hammer ; a mechanical digger; a word processor; a medical diagnostic help system.

Proxies

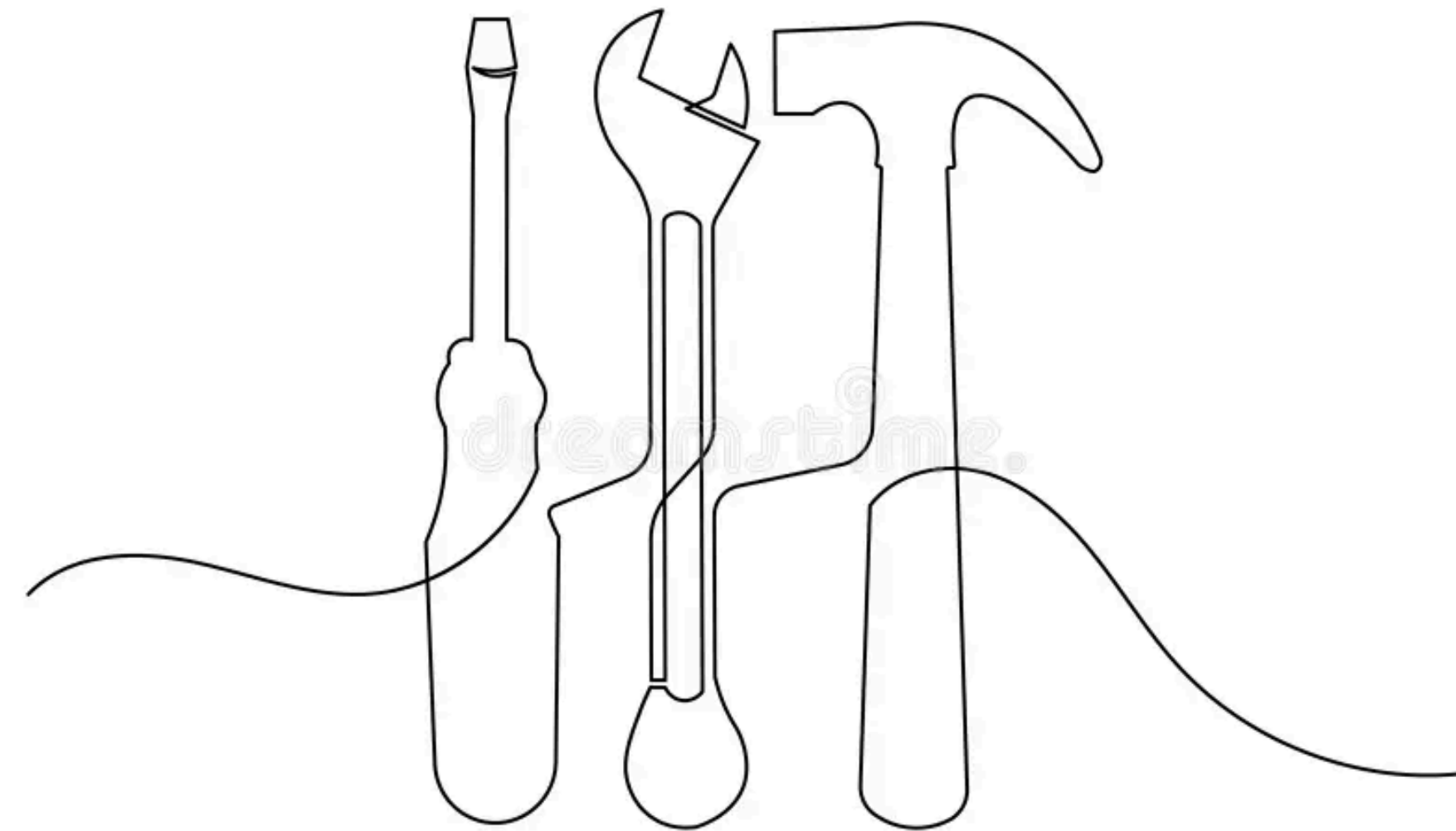
Machines can replace us by doing what we already do perhaps better than we can do it

A rope tied round a bollard to hold a boat still is a proxy; a thermostat; the control belonging to a radar-controlled gun; a computer program that integrates mathematical functions ; an expert system for medical diagnosis i

Novelties

Machines can do types of things that we could never do without them—let us call such machines “novelties

A high bridge; an explosive device; a fridge-freezer; a space rocket; a laser is a novelty; a virtual-reality headset



ANALYSYS LEVEL

When action is broken down into small enough elements, tools become proxies.

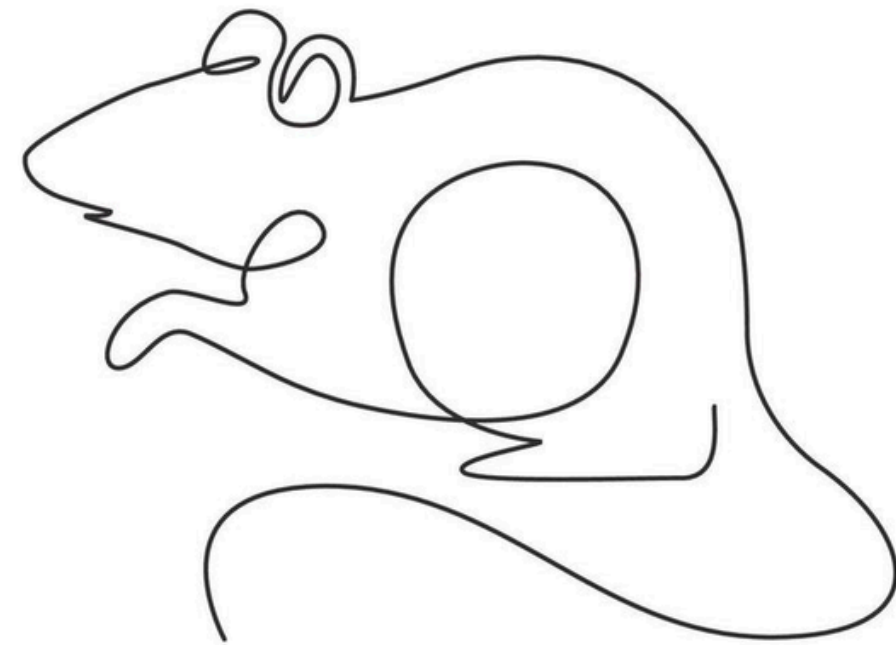


Between the moment that one presses forward the little lever on a tool such as a mechanical digger, to instruct it to drag the bucket teeth first, through the earth toward you, and the moment the bucket comes to the end of its travel, the device is acting not as a tool but as a proxy—it is scooping up earth “by itself.

- On the other hand, we can always turn a proxy into a tool by considering what it does from a vantage point higher up in the action tree

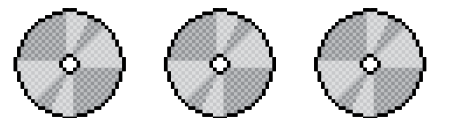
a rope is a tool for our ship-controlling actions, a radar-controlled gun is a tool we use in our war-waging actions, and a word processor is a tool we use in our writing actions.

- The general rule is that we can always turn a tool into a proxy by taking a lower vantage point, and we can always turn a proxy into a tool by taking a higher vantage point.



RAT: HIDING IT, BALANCING IT, KILLING IT

It is easy to mistake tools for proxies because of the ease with which we “repair” the deficiencies of machines and then “attribute” agency to them



RAT - “REPAIR, ATTRIBUTION, AND ALL THAT

Page 07

As três modalidades do RAT

Hiding the RAT: Ocultar do usuário o quanto de trabalho de reparação ele está fazendo. Ex.: interfaces que aproveitam habilidades de reparação já incorporadas na vida cotidiana.

“ $7/11 \times 11$ ” you get the answer 6.9999996, mas você vê 7

Balancing the RAT: Para atribuir corretamente agência de alto nível à máquina, ela precisaria ser tão boa quanto o humano em reparar contextos.

Apenas neste caso poderia passar no Teste de Turing.

Building devices that are as good at RAT as are humans, so that the burden of repair is distributed evenly between both parties in an interaction, even when one is a machine

Killing the RAT: Eliminar a necessidade de reparação restringindo as ações a domínios mimeomórficos. Máquinas podem funcionar como proxies legítimos quando o contexto não requer socialização.



HOW MACHINES WORKS

Page 08

Behavers: Behavers just do the same thing over and over again.

ex: An automatic drilling machine.

Disjunctive behavers: Disjunctive behavers do one of a set choice of things according to a predetermined list of choices.

ex: A chair-spraying robot, programmed to recognize which of a set of types of chair is currently in front of it, and to operate the corresponding program.

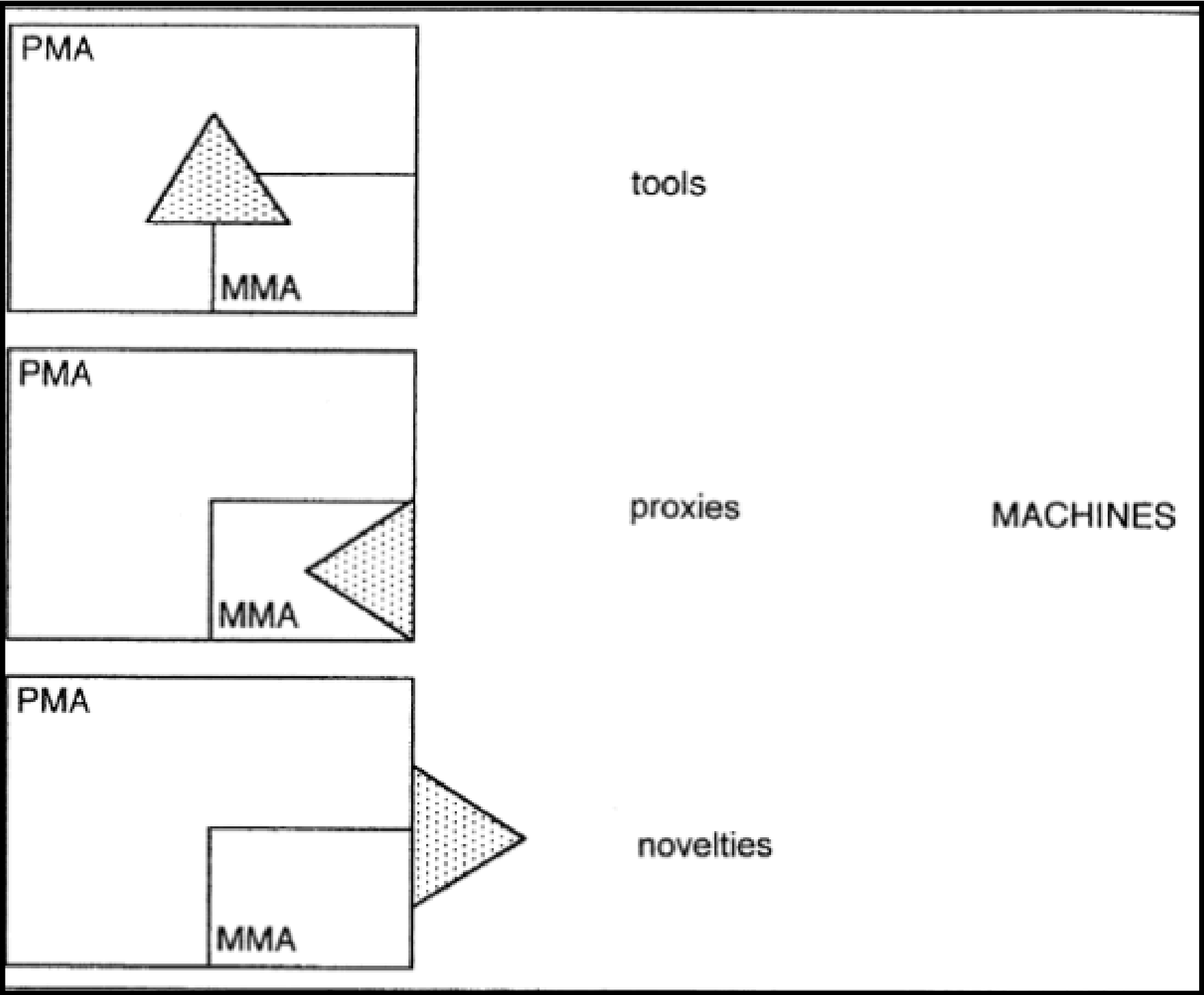
Feedback behavers: Feedback behavers respond not to a fixed set of possibilities but to anything within a preconceived range of sensed stimuli.

ex: A car wash is a feedback behavior

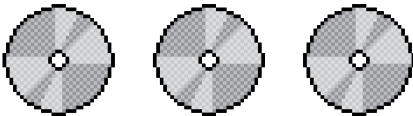
Learning behavers: Learning behavers take in information from their environment and their users, or both, and incorporate this in their design.

Ex: plaster of Paris casts such as are used for holding broken bones in place; A boot

THE TERRITORIES OF MACHINES AND ACTIONS



retângulo PMA : human actions
triângulo : Tools, Proxies e novelties



THE PHYLOGENIC SCALE OF MACHINES



the typology of the ways in which machines work, set out above, if it is a phylogenetic scale, has a much narrower range than first appears. If we can establish the crucial importance of social abilities, we will have shown that even the most advanced neural nets are at the level of the mollusk

INDIFFERENCE MATCHING: THE SOCIOLOGY OF THE SAME

When machines act as proxies in replacing casual mimeomorphic actions, we like it because it saves us time and energy; when machines replace special mimeomorphic actions we like it even more because, usually, we consider that the machines do a better and more accurate job than ourselves. Machine are usually better than us at generating behaviors equivalent to special mimeomorphic actions—good machines are made that way.



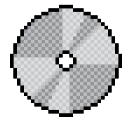
APPENDIX: A ROBOT SLAVE

	morph- icity 1	can be formul- ated 2	route through tree 3	how competence gained 4	level of awareness 5	skill - ful 6	can be mechan- ized 7	how mechan- ized 8
1a semaphor	mimeo	yes	plan	calculation	conscious	no	yes	program
1b pheasant plucker						yes		
2a marching	mimeo	yes	plan or channel	drill	conscious → internalised	no yes	yes	program
2b synchro- swimming								
3 chair- spraying	mimeo	yes	channel	social skill practice	tacit	maybe	yes	research or record and playback
4 bike- balancing in the wild	mimeo	yes or potentially	channel	social skill practice	tacit	yes	yes	research or engineering
5 canoeing down rapids	mimeo	no because too complex	channel	social skill practice	tacit	yes	maybe	engineering
6 bike- riding in ordinary traffic	poli	no because socially embedded	situated action path	social skill	tacit	yes	yes	-----
7 natural speech	poli	no because socially embedded	situated action path	social skill	tacit	no	no	-----

Figure 5.2
Some actions and their properties.

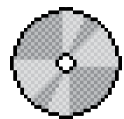
HOW TO BUY A COMPUTER

Match Purpose



A workstation needs specs than a laptop; match performance to tasks.

Check Specs



CPU, memory, storage, and graphics matter; higher numbers often help performance.

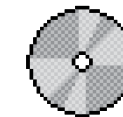
Decide on purpose, set a budget, compare specs, check reviews, and consider upgradeability for future needs and support with extended warranty.





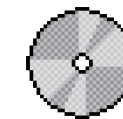
THE FUTURE OF COMPUTING

Smarter Software



Software understands context, assists tasks, making interactions natural and efficient.

Energy Focus



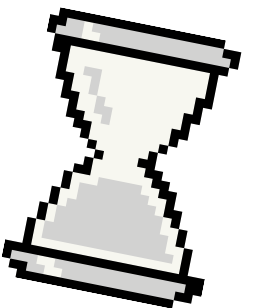
Designs prioritize energy savings, longer battery life, and sustainable materials.

Improvements in speed, energy efficiency, smarter software, and tighter integration with daily life will shape future computers globally.



THANK YOU FOR JOINING US

Computers are powerful, approachable tools; practice regularly, explore responsibly, and enjoy learning new capabilities and possibilities every day.



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